

Appendix for Text Watermarking for Chinese through Homoglyph Substitution

1 Appendix

Table 1: A selection of the single-homoglyph characters and their corresponding hexadecimal codes. The full list of 256 characters is available at <https://drive.google.com/file/d/1nhiskoaG8yBbXrJ3hCPBHZJYpoKoJbAi/view>.

Character	Original	Homoglyph	Character	Original	Homoglyph
八	0x516B	0x2F0B	白	0x767D	0x2F69
暴	0x66B4	0xFA06	北	0x5317	0xF963
贝	0x8D1D	0x2EC9	卑	0x5351	0xFA35
碑	0x7891	0xFA4B	鼻	0x9F3B	0x2FD0
匕	0x5315	0x2F14	便	0x4FBF	0xF965
比	0x6BD4	0x2F50	丙	0x4E19	0x319B
不	0x4E0D	0xF967	卜	0x535C	0x2F18
茶	0x8336	0xF9FE	长	0x957F	0x2ED3
厂	0x5382	0x2F1A	车	0x8F66	0x2ECB
臣	0x81E3	0x2F82	齿	0x9F7F	0x2EEE
赤	0x8D64	0x2F9A	虫	0x866B	0x2F8D

Table 2: All 17 double-homoglyph characters with their respective hexadecimal codes.

Character	Original	I Homoglyph	II Homolgyph
辰	0x8FB0	0xF971	0x2FA0
二	0x4E8C	0x2F06	0x3193
金	0x91D1	0xF90A	0x2FA6
口	0x53E3	0x2F1D	0x2F1E
老	0x8001	0xF934	0x2F7C
立	0x7ACB	0xF9F7	0x2F74
里	0x91CC	0xF9E9	0x2FA5
力	0x529B	0xF98A	0x2F12
鹿	0x9E7F	0xF940	0x2FC5
率	0x7387	0xF9DB	0xF961
女	0x5973	0xF981	0x2F25
人	0x4EBA	0x2F08	0x319F
食	0x98DF	0x2EDD	0x2FB7
兀	0x5140	0x2E8E	0xFA0C
行	0x884C	0xFA08	0x2F8F
一	0x4E00	0x2F00	0x3192
乙	0x4E59	0x2F04	0x319A

Table 3: All 14 Latin characters with their respective hexadecimal codes.

Character	Original	Homoglyph
C	0x0043	0x216d
c	0x0063	0x217d
D	0x0044	0x216e
d	0x0064	0x217e
i	0x0069	0x2170
j	0x006a	0x0458
K	0x004b	0x212a
L	0x004c	0x216c
l	0x006c	0x217c
M	0x004d	0x216f
V	0x0056	0x2164
v	0x0076	0x2174
X	0x0058	0x2169
x	0x0078	0x2179

Table 4: The twelve characters that are rendered as superscripts on certain platforms (*i.e.*, Windows and Apple) and in specific fonts (*i.e.*, *Arial*, *Microsoft YaHei*, and *SimSun*).

Character	Original	Homoglyph
丙	0x4E19	0x319B
地	0x5730	0x319E
丁	0x4E01	0x319C
二	0x4E8C	0x3193
甲	0x7532	0x3199
人	0x4EBA	0x319F
三	0x4E09	0x3194
上	0x4E0A	0x3196
四	0x56DB	0x3195
一	0x4E00	0x3192
乙	0x4E59	0x319A
中	0x4E2D	0x3197

Algorithm 1: Watermarking embedding, with repeated embedding.

```
1 /* Text  $c$  composed of  $n$  characters */
2  $T \leftarrow \{c_1, \dots, c_n\}$ ;
3 /* List of originals with one or two duplicates */
4  $OriginalsSingle \leftarrow \{0x516B, 0x66B4, \dots\}$ ;
5  $OriginalsDouble \leftarrow \{0x8FB0, 0x4E8C, \dots\}$ ;
6  $Punctuations \leftarrow \{0x002C, 0x003B, \dots\}$ ;
7 /* List of single and double duplicates */
8  $SingleDuplicates \leftarrow \{0x516B, 0x66B4, \dots\}$ ;
9  $DoubleDuplicates \leftarrow \{0x8FB0, 0x4E8C, \dots\}$ ;
10  $Spaces \leftarrow \{0x0020, 0x2000, \dots\}$ ;
11  $Confusables \leftarrow SingleDuplicates \cup DoubleDuplicates \cup Spaces$ ;
12  $WT \leftarrow []$ ;
13  $GetSingleDuplicate : OriginalsSingle \rightarrow SingleDuplicates$ ;
14  $GetDoubleDuplicate : (OriginalsDouble, \{0, 1\}) \rightarrow DoubleDuplicates$ ;
15  $GetSpace : \{000, \dots, 111\} \rightarrow Spaces$ ;
16  $i = 0$ ; /* Pointer to watermark index */
17 foreach  $c \in T$  do
18   if  $c \in Confusables$  then
19     if  $c \in Spaces$  then
20        $bits = []$ ;
21       for  $ii = 0..2$  do
22          $Append(bits, W[(i + ii) \% 64])$ ;
23        $c \leftarrow GetSpace(bits)$ ;
24     else if  $c \in Punctuations$  then
25        $bits1 = []$ ;
26       for  $ii = 0..2$  do
27          $Append(bits1, W[(i + ii) \% 64])$ ;
28        $bits2 = []$ ;
29       for  $ii = 0..2$  do
30          $Append(bits2, W[(i + ii) \% 64])$ ;
31        $c \leftarrow GetSingleDuplicate(c) + GetSpaces(bits1) +$ 
32          $GetSpaces(bits2)$ ;
33     else
34        $bit = W[i \% 64]$ ;
35       if  $bit = 1$  then
36         if  $c \in DoubleDuplicates$  then
37           if  $\neg Eof(W)$  then
38              $bit2 = W[i \% 64]$ ;
39           else
40              $bit2 = W[0]$ ;
41            $c \leftarrow GetDoubleDuplicate(c, bit2)$ ;
42         else
43            $c \leftarrow GetSingleDuplicate(c)$ ;
44    $Append(WT, c)$ ;
45 return  $WT$ 
```

Algorithm 2: Watermark extraction.

```
1 /* Text string  $c$  composed of  $n$  characters */
2  $T \leftarrow \{c_1, \dots, c_n\}$ ;
3 /* List of originals with one or two duplicates */
4  $Originals \leftarrow \{0x516B, 0x66B4, \dots\}$ ;
5  $Punctuations \leftarrow \{0x002C, 0x003B, \dots\}$ ;
6 /* Lists of single and double duplicates */
7  $SingleDuplicates \leftarrow \{0x516B, 0x66B4, \dots\}$ ;
8  $DoubleDuplicates \leftarrow \{0x8FB0, 0x4E8C, \dots\}$ ;
9  $Spaces \leftarrow \{0x0020, 0x2000, \dots\}$ ;
10 /* Watermarking sequence */
11  $W \leftarrow []$ 
12 /* GetSpace inverse function */
13  $GetSpaceBit : Spaces \rightarrow \{000, \dots, 111\}$ 
14  $GetDoubleDupBit : DoubleDuplicates \rightarrow \{10, 11\}$ 
15 foreach  $c \in T$  do
16   if  $c \in Punctuations$  then
17     continue
18   else if  $c \in Spaces$  then
19      $Append(W, GetSpaceBit(c))$ ;
20   else if  $c \in DoubleDuplicates$  then
21      $Append(W, GetDoubleDupBit(c))$ ;
22   else if  $c \in SingleDuplicates$  then
23      $Append(W, 1)$ ;
24   else if  $c \in Originals$  then
25      $Append(W, 0)$ ;
26   if  $|W| \geq 64$  then
27     break
28 return  $W[: 64]$ 
```
